

Section 1: Heart Failure Classification and Diagnosis

Heart failure is categorized by Left Ventricular Ejection Fraction (LVEF): Heart Failure with Reduced Ejection Fraction (HFrEF: LVEF $\leq$ 40%), Heart Failure with Mildly Reduced Ejection Fraction (HFmrEF: LVEF 41-49%), and Heart Failure with Preserved Ejection Fraction (HFpEF: LVEF $\geq$ 50%).

New York Heart Association (NYHA) Functional Classes: Class I (No limitation of physical activity), Class II (Slight limitation, comfortable at rest), Class III (Marked limitation of physical activity), Class IV (Unable to carry on any physical activity without discomfort; symptoms at rest).

Section 2: Guideline-Directed Medical Therapy (GDMT) for HFrEF

Foundational GDMT consists of Quadruple Therapy to reduce morbidity and mortality:

1. Angiotensin Receptor-Neprilysin Inhibitor (ARNI, Sacubitril/Valsartan) preferred over ACE inhibitors or ARBs.
2. Evidence-Based Beta-Blockers: Bisoprolol, Carvedilol, or sustained-release Metoprolol Succinate.
3. Mineralocorticoid Receptor Antagonists (MRA): Spironolactone or Eplerenone (monitor serum potassium and renal function).
4. SGLT2 Inhibitors: Dapagliflozin (10 mg daily) or Empagliflozin (10 mg daily), recommended for all symptomatic HFrEF patients regardless of diabetes status.

Section 3: Acute Coronary Syndrome (ACS) Management

Immediate medical therapy for Acute Coronary Syndrome includes chewable Aspirin (162-325 mg), sublingual Nitroglycerin for ischemic pain, and high-intensity Statin therapy (Atorvastatin 80 mg or Rosuvastatin 40 mg).

Dual Antiplatelet Therapy (DAPT): Combine Aspirin with a P2Y₁₂ inhibitor (Ticagrelor 90 mg twice daily or Prasugrel 10 mg once daily preferred over Clopidogrel) for at least 12 months following percutaneous coronary intervention (PCI).